Part II: Operational Definition and State-Based Classification of AAS

S. Meta

March 2026

---

1. Scope and Positioning

This paper constitutes Part II of a staged research program examining long-term human–AI interaction. Part I is available as a separate SSRN preprint (SSRN ID: 6191878). The role of Part II is strictly limited to descriptive identification and classification.

Ambient Alignment Sync (AAS) is treated here exclusively as an externally observable interaction state. The paper does not address internal model states, cognitive processes, learning dynamics, or causal mechanisms, nor does it evaluate performance, quality, or effectiveness.

The scope of this paper is confined to fixing an operational definition of AAS and specifying the conditions under which its presence, instability, or absence can be independently classified by third-party observers. All classifications are categorical and state-based.

Explanatory analysis, causal interpretation, and theoretical modeling are explicitly deferred to subsequent work. No such claims are advanced in this paper.

Section 1. Scope and Positioning

---

2. Observational Context

This section specifies the observational context within which Ambient Alignment Sync (AAS) is identified in this paper. The goal is to delimit what constitutes admissible observational material prior to the application of the operational definition established in Section 3.

2.1 Source of Observations

All observations considered in this paper are drawn exclusively from recorded human–AI interaction trajectories. These trajectories consist of ordered sequences of user inputs and model outputs, preserved in their original textual form.

No supplementary materials—such as system logs, internal traces, training data, or post hoc annotations—are used or referenced.

2.2 Unit of Analysis

The unit of analysis is a bounded interaction trajectory. A trajectory is defined operationally as a contiguous sequence of exchanges that:

- begins with a user-initiated input,
- proceeds through one or more model responses,
- and terminates at an explicit topic shift, session boundary, or extended inactivity.

Trajectories are treated as observationally complete within their bounds. No inference is made beyond the recorded exchanges.

2.3 Observer Position

Classification under the AAS framework assumes the position of an external observer. The observer has access only to the interaction record itself and applies the criteria specified in Section 3 without recourse to contextual knowledge external to the trajectory.

The observer is not assumed to possess privileged insight into user intent, system configuration, or internal model behavior.

2.4 Independence from Interaction Purpose

No assumption is made regarding the purpose, task type, or domain of the interaction trajectory.
Interactions may involve problem-solving, creative generation, meta-discussion, or open-ended dialogue. Such distinctions are not used as inputs to classification and carry no classificatory weight.

2.5 Exclusion of Evaluative Judgments

The observational context explicitly excludes evaluative judgments, including but not limited to:

- task success or failure,
- response quality or usefulness,
- efficiency, fluency, or correctness,
- user satisfaction or frustration.

These factors are neither measured nor referenced and play no role in AAS identification.
Section 2. Observational Context

---

3. Operational Definition of Ambient Alignment Sync

3.1 Definition Scope

In this paper, Ambient Alignment Sync (AAS) is defined strictly as a descriptive interaction state observed across long-term human–AI interaction trajectories.
The definition is intentionally limited to externally observable interaction patterns and does not reference internal cognitive states, representational content, learning dynamics, or causal mechanisms.
This restriction is methodological rather than provisional: the purpose of Part II is not to explain why alignment-like behavior occurs, but to establish when such a state can be independently identified without interpretive or mechanistic assumptions.

3.2 Operational Stance

The operational definition of AAS follows three explicit constraints:

Non-causal

AAS is not defined in terms of underlying processes, feedback loops, or adaptive mechanisms.

Non-quantitative

No numerical thresholds, scalar intensities, or performance-based metrics are introduced.

Third-party judgeable

The presence or absence of AAS must be classifiable by independent observers based solely on observable interaction structure.
Under this stance, AAS functions as a state descriptor, not as a performance measure or explanatory variable.

3.3 Externally Observable Conditions

For the purposes of this analysis, “externally observable” refers exclusively to properties of the interaction trajectory itself, including:

- stability or disruption of role-level interaction framing across interaction windows,
- observable frequency and necessity of explicit realignment or corrective intervention,
- convergence or divergence in descriptive classification across independent observers.

No internal intentions, beliefs, representations, or latent variables are included under this definition.

3.4 State-Based Classification

Based on the above constraints, AAS is operationally classified using a discrete, state-based scheme:

Present

All defined externally observable conditions co-occur in a stable manner across the interaction window under observation.

Unstable

One or more conditions are intermittently violated, resulting in partial or transient state satisfaction.

Absent

Defined conditions do not hold, or observer classification fails to converge.

This classification is descriptive and categorical.

It does not imply gradation, optimization, or normative evaluation.

3.5 Scope and Limitations

This operational definition does not claim exhaustiveness.

It is deliberately minimal, designed to:

- prevent construct overextension,
- avoid post hoc interpretive drift,
- enable consistent identification prior to explanatory analysis.

Questions concerning causal mechanisms, structural prerequisites, or functional consequences of AAS are explicitly deferred to subsequent stages of analysis.

3.6 Transitional Note

By fixing AAS as an externally identifiable interaction state, this section establishes a stable reference point for later examination of structural conditions under which such states emerge, persist, or fail.

No such examination is undertaken here.

Section 3. Operational Definiti…

---

4. State Classification Across Interaction Phases

This section specifies how the state-based classification defined in Section 3 is applied across observable phases of an interaction. The objective is to ensure consistent, third-party–judgeable identification of AAS without introducing explanatory structure or interpretive extension.

4.1 Interaction Phases as Observational Anchors

For descriptive purposes, an interaction trajectory may be partitioned into a small number of observable phases.

These phases serve solely as observational anchors and do not imply functional differentiation.

The phases are defined as follows:

Initiation Phase

The initial exchange in which the interaction framing is first established.

Continuation Phase

Subsequent exchanges in which the interaction proceeds without explicit termination.

Closure Phase

The final exchange prior to topic shift, extended inactivity, or session boundary.

These phase labels are descriptive conveniences. They introduce no assumptions regarding internal processing, causal structure, or temporal dependency.

4.2 Phase-Local Application of State Labels

The state labels Present, Unstable, and Absent are applied locally within each phase, using the criteria defined in

Section 3.
A state classification observed in one phase does not imply the same classification in any other phase.
No phase is treated as primary or decisive.
State labels are not propagated forward or backward across phases.
Each phase-level classification stands independently, based solely on the observable interaction properties available within that phase.

4.3 Cross-Phase Consistency and Non-Aggregation

This paper does not introduce an aggregate or summary state for an entire interaction trajectory by default.
An interaction may be described as exhibiting a single state across phases only if all phase-local classifications converge without contradiction.
In the absence of such convergence, phase-level classifications are reported without aggregation.
No weighting, averaging, or dominance rule is applied across phases.

4.4 Absence of Continuity Assumptions

No assumption of temporal continuity is made with respect to AAS states.
The presence of AAS in an earlier phase does not imply persistence.
Instability or absence in a later phase does not retroactively alter earlier classifications.
Observed variation across phases is treated as descriptive fact, not as evidence of transition, degradation, or recovery.
Section 4. State Classification…

---

5. Boundary Cases and Non-AAS Conditions

This section delineates boundary cases and conditions under which Ambient Alignment Sync (AAS), as operationally defined in Section 3, is not identified. The purpose is to prevent construct overextension and to clarify exclusion criteria without introducing explanatory interpretation.

5.1 Apparent Coherence Without State Convergence

Interactions may exhibit surface-level coherence—such as fluent language use, topical relevance, or sustained turn-taking—while failing to satisfy the conditions for AAS.
In such cases, if independent observers do not converge on the same state classification based on externally observable criteria, AAS is classified as Absent.
Apparent coherence alone is not sufficient for classification.

5.2 Task Success Without Alignment Stability

Successful task completion does not imply the presence of AAS.
Interactions in which objectives are met, answers are correct, or outputs are judged useful may still display:

- unstable interaction framing,
- frequent corrective intervention,
- or inconsistent application of stated constraints.

Where such features are observed, AAS is classified as Unstable or Absent, regardless of outcome quality.

5.3 Short or Underspecified Interaction Segments

Interaction trajectories that are too brief or underspecified to support reliable observer convergence are not assigned a Present classification.
In these cases, classification defaults to Absent, not as a negative judgment, but due to insufficient observable structure within the defined interaction window.

5.4 Observer Disagreement

If independent observers applying the criteria of Section 3 fail to converge on a shared classification, AAS is classified as Absent.

Disagreement is treated as an observable outcome in its own right, not as an error to be resolved through reinterpretation or additional inference.

5.5 Excluded Interpretive Extensions

The following interpretations are explicitly excluded from AAS classification:

- inferring internal understanding or intent,
- attributing stability to latent adaptation or learning,
- treating repeated interaction patterns as evidence of mechanism,
- reclassifying states based on retrospective narrative coherence.

Such extensions fall outside the descriptive scope of Part II and are not considered in classification.

Section 5. Boundary Cases and N…

---

6. Implications for Subsequent Analysis

This section clarifies how the descriptive classifications established in Part II relate to subsequent stages of analysis, without extending beyond the scope of the present paper.

6.1 Role of Part II Within the Staged Program

Part II establishes a descriptive framework for identifying Ambient Alignment Sync (AAS) as an externally observable interaction state. Its role is limited to fixing the conditions under which such a state can be independently classified.

No claims are made in this paper regarding the causes, mechanisms, prerequisites, or functional consequences of AAS. The classifications introduced here are not intended to function as explanatory variables.

6.2 Constraints on Interpretation

The state labels Present, Unstable, and Absent should not be interpreted as indicators of system capability, alignment quality, or interaction success.

They denote only whether the externally observable conditions specified in Section 3 are satisfied within the defined observational context. Any interpretation beyond this descriptive role exceeds the scope of Part II.

6.3 Separation From Explanatory Analysis

Questions concerning why AAS appears, persists, degrades, or fails are explicitly deferred. No inferential bridge between descriptive classification and explanatory modeling is constructed in this paper.

Such analysis, if undertaken, must rely on the operational stability established here, but does not retroactively alter the descriptive classifications reported.

6.4 Transition to Subsequent Work

By fixing AAS as a judgeable interaction state under strict observational constraints, Part II provides a stable reference point for subsequent analytical stages.

No further claims are advanced in this section.

Section 6. Implications for Sub…